

ElectroHub E-Commerce Analytics

Industry-Style Business Problem & Case Study Practice

Objective: Practice thinking like a Data Analyst — start with a business problem, identify the required data, analyze drivers, and communicate evidence-based findings.

1. Business Context

ElectroHub is an e-commerce business that wants to improve sales and profit while retaining valuable customers and reducing the business impact of delivery problems, cancellations, returns, and inefficient discounting.

Core Business Problem: How can ElectroHub increase sales and profit while retaining valuable customers and reducing the business impact of poor delivery, cancellations, returns, and inefficient discounting?

2. Available Data

Table	Important Columns	Business Role
customers	Customer_ID, State, Customer_Tier, Acquisition_Channel, Age, Gender	Customer profile and acquisition
orders	Order_ID, Customer_ID, Order_Date	Transactions and order timing
order_items	Order_ID, Product_ID, Quantity, Unit_Price, Gross_Sales, Discount_Pct, Discount_Amount, Net_Sales, Cost, Profit	Product-level financial details
products	Product_ID, Product_Name, Category	Product and category information
delivery	Order_ID, Warehouse, Promised_Days, Actual_Delivery_Days, Shipping_Cost, Status, Payment_Method	Delivery and operational performance

3. Management Questions

A. Overall Performance

1. What are the total sales, profit, orders, customers, AOV, and profit margin?
2. How has sales changed over time?
3. Is sales growth translating into profitable growth?

B. Product & Profitability

4. Which categories drive the most sales?
5. Which categories drive the most profit?
6. Which products have high sales but relatively low profit?
7. How might discounting be affecting profit and margin?

C. Customer & Growth

8. Which customer tiers contribute the most sales?
9. Which acquisition channels generate the most customers and sales?
10. Which channels generate higher-value orders/AOV?
11. What is the repeat-customer level, and why should management monitor it?

D. Operations & Delivery

12. What is the average actual delivery time?
13. What percentage of orders are delivered later than promised?
14. Which warehouses or states show weaker delivery performance?
15. How should cancellations, returns, and shipping cost be monitored alongside delivery performance?

4. Expected Answer Structure

Step	What to write
Business Question	What decision/problem are we trying to understand?
Metric	Which KPI measures the problem?
Analysis	What SQL/Python calculation or breakdown did you perform?
Finding	What does the data show? Include the actual value from your analysis.
Driver	Which category, product, state, customer group, channel, etc. contributed?
Business Impact	Why does this matter to sales, profit, customers, or operations?
Recommendation	What should management investigate or consider doing?
Monitoring	Which KPI should be tracked after action?

5. Important Business Rules

- High sales do not automatically mean high profitability.
- Customer count alone does not measure customer value.
- A high acquisition volume does not automatically mean a high-value channel.
- A late delivery can be a genuine operational event and should not simply be deleted as an outlier.
- Negative discounts are suspicious/invalid under the project's intended business logic and should be investigated.
- Recommendations should be based on evidence, not only on a chart.